In [ ]:
```
# ================================
# Step 1: Load CSV into DataFrame
# ================================

from google.colab import files
uploaded = files.upload()
```

Upload widget is only available when the cell has been executed in the current browser session.
Please rerun this cell to enable.

```
Saving hotel_bookings.csv to hotel_bookings (4).csv
```

In [ ]:
```
import pandas as pd

df = pd.read_csv("hotel_bookings.csv")

# Optional preview
df.head()
```

Out[ ]:

	hotel	is_canceled	lead_time	arrival_date_year	arrival_date_month	arrival_date_we
0	Resort Hotel	0	342	2015	July	
1	Resort Hotel	0	737	2015	July	
2	Resort Hotel	0	7	2015	July	
3	Resort Hotel	0	13	2015	July	
4	Resort Hotel	0	14	2015	July	

5 rows × 32 columns

In [ ]:
```
# ======================================
# Step 2: Initial Data Audit and Preview
# ======================================

# Check dataset structure
print("Shape:", df.shape)
print("\nData types:")
print(df.dtypes)

# Check for missing values
print("\nMissing values per column:")
print(df.isnull().sum().sort_values(ascending=False))

# View unique values for key categorical columns
categorical_cols = ['hotel', 'market_segment', 'customer_type', 'depo
sit_type', 'distribution_channel']
for col in categorical_cols:
    print(f"\nUnique values in '{col}':")
    print(df[col].value_counts())
```

```
Shape: (119390, 32)

Data types:
hotel                               object
is_canceled                          int64
lead_time                            int64
arrival_date_year                    int64
arrival_date_month                  object
arrival_date_week_number             int64
arrival_date_day_of_month            int64
stays_in_weekend_nights              int64
stays_in_week_nights                 int64
adults                               int64
children                           float64
babies                               int64
meal                                object
country                             object
market_segment                      object
distribution_channel                object
is_repeated_guest                    int64
previous_cancellations               int64
previous_bookings_not_canceled       int64
reserved_room_type                  object
assigned_room_type                  object
booking_changes                      int64
deposit_type                        object
agent                              float64
company                            float64
days_in_waiting_list                 int64
customer_type                       object
adr                                float64
required_car_parking_spaces          int64
total_of_special_requests            int64
reservation_status                  object
reservation_status_date             object
dtype: object

Missing values per column:
company                            112593
agent                               16340
country                               488
children                                4
arrival_date_month                      0
arrival_date_week_number                0
hotel                                   0
is_canceled                             0
stays_in_weekend_nights                 0
arrival_date_day_of_month               0
adults                                  0
stays_in_week_nights                    0
babies                                  0
meal                                    0
lead_time                               0
arrival_date_year                       0
distribution_channel                    0
market_segment                          0
previous_bookings_not_canceled          0
is_repeated_guest                       0
reserved_room_type                      0
assigned_room_type                      0
booking_changes                         0
```

In [ ]:
```
#high missingness but are not needed for your current experimental
design:
#company (94% missing): optional ID field, safe to ignore
#agent (13% missing): can be excluded unless you're analyzing booking
intermediaries
#country: low missingness (488 rows), could be dropped or imputed if
needed
#children: only 4 missing values — could drop or fill with 0
```

In [ ]:
```
# Drop rows with missing 'children'
df = df[df['children'].notna()]

# Optionally drop irrelevant columns
df = df.drop(columns=['company', 'agent'])
```

In [ ]:
```
df['market_segment'].value_counts()
```

Out[ ]:

	count
market_segment	
Online TA	56476
Offline TA/TO	24219
Groups	19811
Direct	12605
Corporate	5295
Complementary	743
Aviation	237

dtype: int64

In [ ]:
```
# ====================================
# Step 3: Define Treatment (Booking Channel)
# ====================================

# Drop undefined market_segment rows (if any)
df = df[df['market_segment'] != 'Undefined']


df['market_segment'].value_counts()

# Create new binary treatment group based on booking channel
df['booking_channel'] = df['market_segment'].apply(
    lambda x: 'Online' if x == 'Online TA' else 'Offline'
)

# Check result
df['booking_channel'].value_counts()
```

Out[ ]:

booking_channel	count
Offline	62910
Online	56476

dtype: int64

In [ ]:
```
df[['market_segment', 'booking_channel']].head(10)
```

Out[ ]:

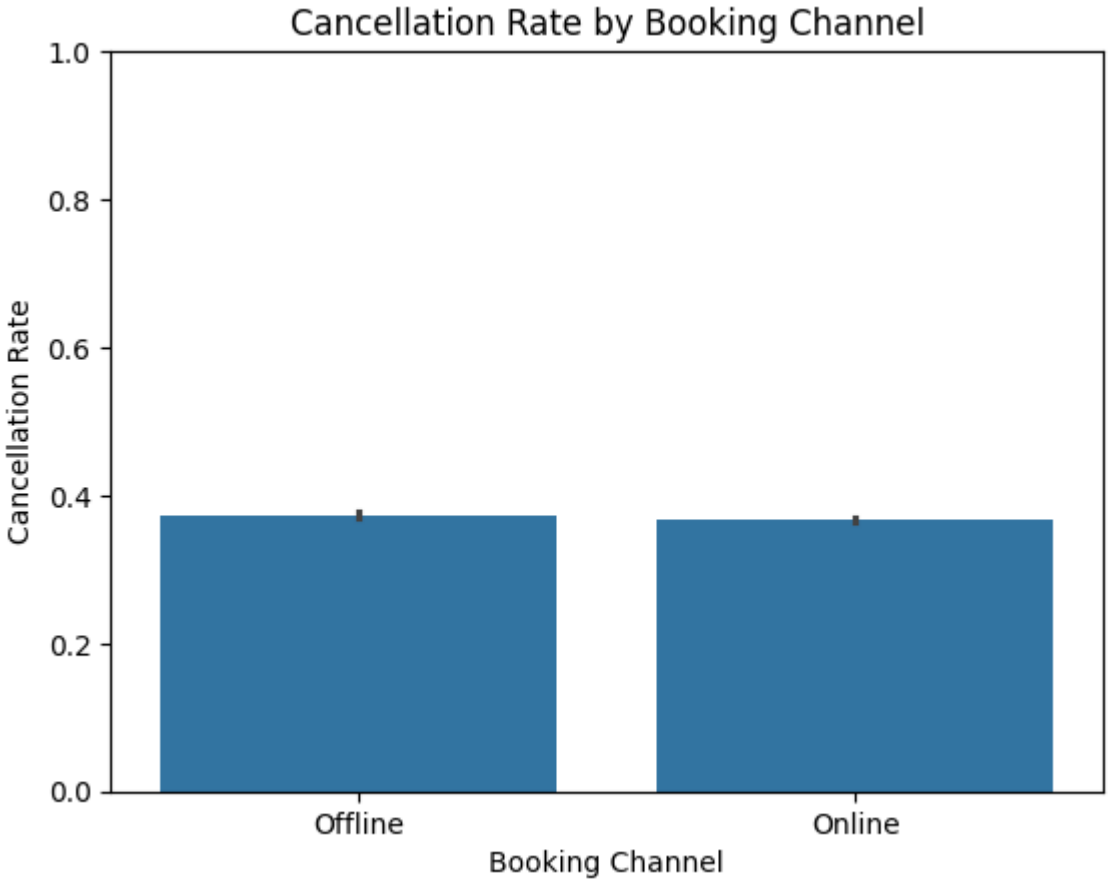
	market_segment	booking_channel
0	Direct	Offline
1	Direct	Offline
2	Direct	Offline
3	Corporate	Offline
4	Online TA	Online
5	Online TA	Online
6	Direct	Offline
7	Direct	Offline
8	Online TA	Online
9	Offline TA/TO	Offline

In [ ]:
```
# ============================================================
# Step 4: Compare Cancellation Rates by Booking Channel
# ============================================================


#Cancellation rate = mean of is_canceled (1 = canceled) grouped by
booking channel

cancellation_rates = df.groupby('booking_channel')['is_canceled'].agg(
['mean', 'count'])
cancellation_rates.rename(columns={'mean': 'cancellation_rate', 'count
': 'n'}, inplace=True)
display(cancellation_rates)

#This gives:
#Mean = cancellation rate (proportion of bookings canceled)
#Count = number of bookings in each group
```

booking_channel	cancellation_rate	n
Offline	0.373263	62910
Online	0.367200	56476

In [ ]:
```
# ==============================================
# Step 4B: Visualize Cancellation Rate (Barplot)
# ==============================================

import seaborn as sns
import matplotlib.pyplot as plt

# Barplot showing mean cancellation rate with 95% CI
sns.barplot(x='booking_channel', y='is_canceled', data=df, estimator='
mean', ci=95)
plt.title('Cancellation Rate by Booking Channel')
plt.ylabel('Cancellation Rate')
plt.xlabel('Booking Channel')
plt.ylim(0, 1)
plt.show()
```

```
<ipython-input-69-6c38f8016706>:9: FutureWarning:

The `ci` parameter is deprecated. Use `errorbar=('ci', 95)` for the same eff

  sns.barplot(x='booking_channel', y='is_canceled', data=df, estimator='mean
ci=95)
```



In [ ]:
```
# ========================================================
# Step 4C: Updated Cancellation Rates by Booking Channel
# ========================================================

# Group by booking_channel and calculate cancellation rate and count
cancellation_rates = df.groupby('booking_channel')['is_canceled'].agg(
['mean', 'count'])
cancellation_rates.rename(columns={'mean': 'cancellation_rate', 'count
': 'n'}, inplace=True)
display(cancellation_rates)

# --------------------------------------------------------
# This gives the current cancellation rate per group
# after cleaning and reclassifying booking_channel.
# We'll use these counts in the z-test next.
# --------------------------------------------------------
```

	cancellation_rate	n
booking_channel		
Offline	0.373263	62910
Online	0.367200	56476

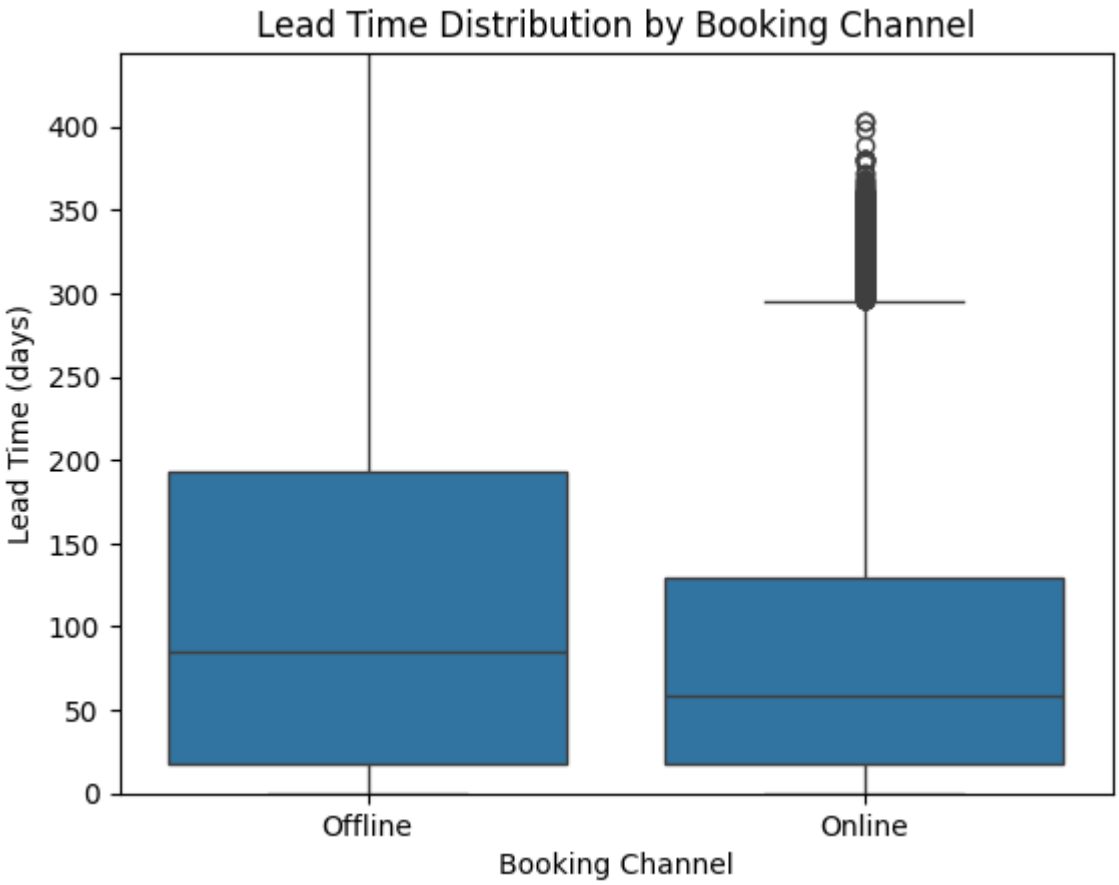
In [ ]:
```
# ========================================================
# Step 4D: Two-Proportion Z-Test for Cancellation Rates
# ========================================================

from statsmodels.stats.proportion import proportions_ztest

# Count cancellations in each group
cancel_counts = [
    df[(df['booking_channel'] == 'Online') & (df['is_canceled'] ==
1)].shape[0],
    df[(df['booking_channel'] == 'Offline') & (df['is_canceled'] == 1)
].shape[0]
]

# Total observations per group
n_obs = [
    df[df['booking_channel'] == 'Online'].shape[0],
    df[df['booking_channel'] == 'Offline'].shape[0]
]

# Perform two-proportion z-test
z_stat, p_val = proportions_ztest(cancel_counts, n_obs)

# Print result
print(f"Z-statistic: {z_stat:.4f}")
print(f"P-value: {p_val:.4f}")

# --------------------------------------------------------
# Interpretation:
# This tests the null hypothesis that the cancellation
# rates are the same in Online and Offline groups.
# It does NOT control for lead time, deposit type, etc.
# --------------------------------------------------------
```

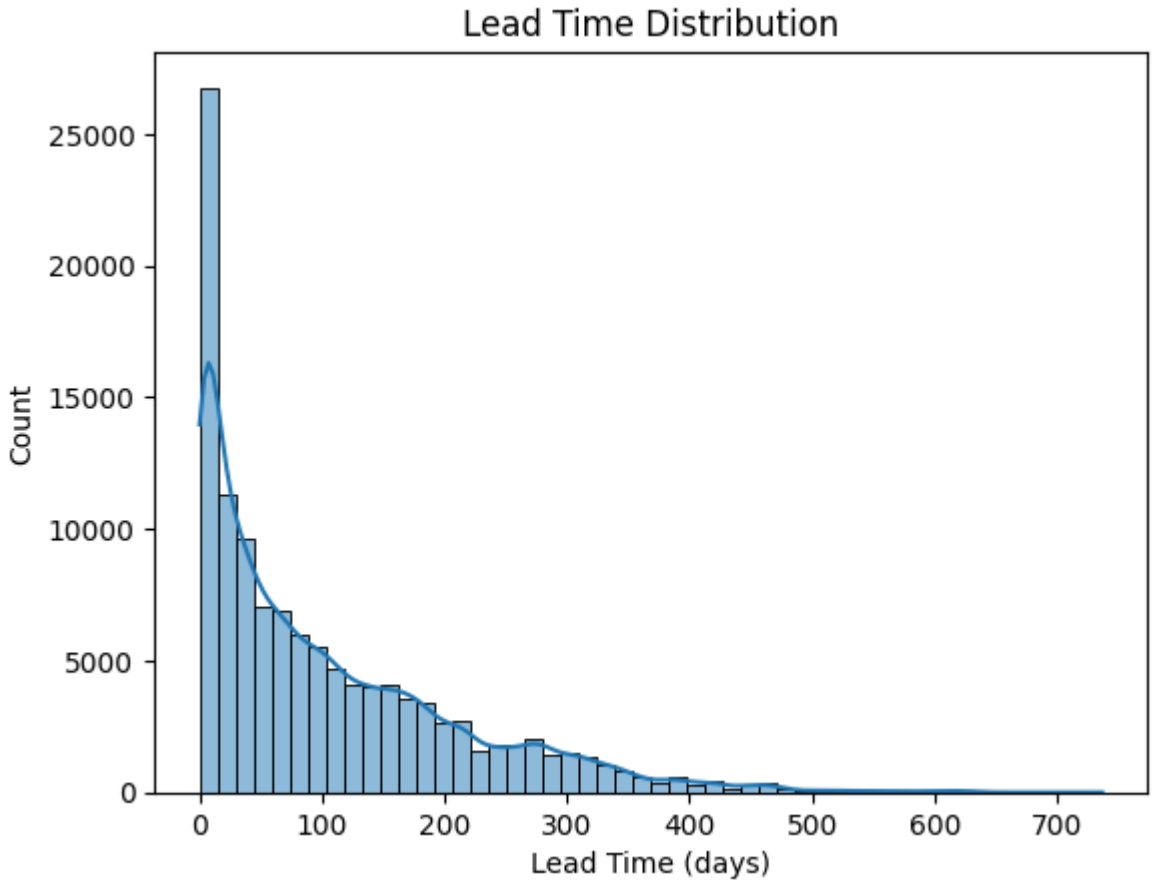
```
Z-statistic: -2.1659
P-value: 0.0303
```

In [ ]:
```
#A two-proportion z-test was conducted to compare cancellation rates
between Online and Offline booking channels. The null hypothesis
assumes equal cancellation rates.

#The test yielded a Z-statistic of −2.17 and a p-value of 0.0303,
indicating a statistically significant difference in cancellation
rates at the 5% level (α = 0.05).
#Specifically, the Offline group had a slightly higher cancellation
rate (37.33%) compared to the Online group (36.72%).

#However, this unadjusted test does not account for confounding
variables such as lead time, deposit type, or customer profile.
#We therefore proceed to a covariate-adjusted logistic regression
model to better isolate the true effect of booking channel on
cancellation behavior.
```

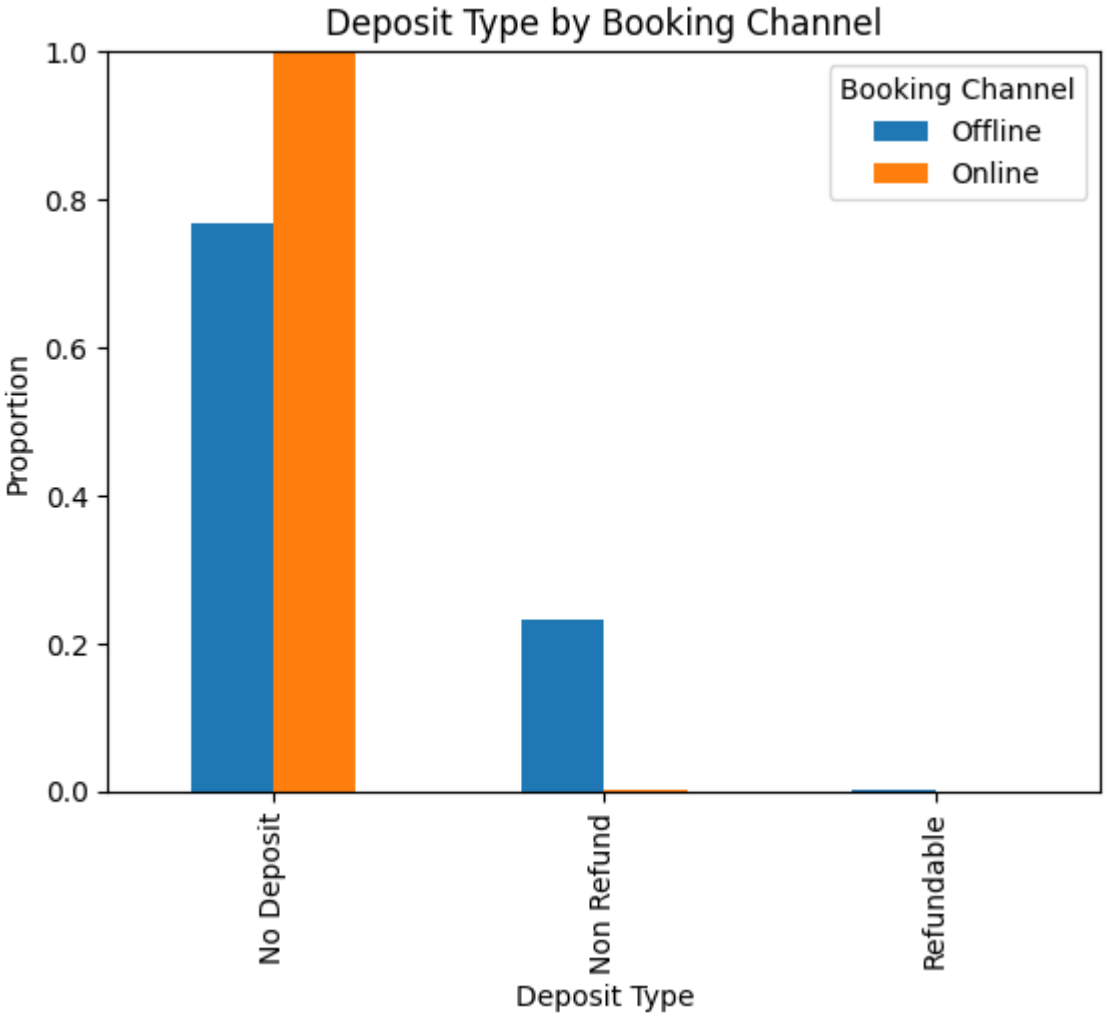
In [ ]:

```python
# ==============================================
# Step 5: Covariate Balance — Lead Time Check
# ==============================================

import seaborn as sns
import matplotlib.pyplot as plt

# Boxplot: Compare lead_time between Online and Offline groups
sns.boxplot(x='booking_channel', y='lead_time', data=df)

# Add title and axis labels
plt.title('Lead Time Distribution by Booking Channel')
plt.ylabel('Lead Time (days)')
plt.xlabel('Booking Channel')

# Limit y-axis to reduce visual skew from extreme outliers
plt.ylim(0, df['lead_time'].quantile(0.99))

# Display the plot
plt.show()

# -------------------------------------------------------
# Interpretation:
# Offline bookings have longer lead times on average,
# and their distribution is much more spread out.
# This imbalance suggests lead_time could confound
# the relationship between booking_channel and is_canceled.
# We'll include it as a covariate in our regression model.
# -------------------------------------------------------



# Visualize lead_time distribution
import seaborn as sns
sns.histplot(df['lead_time'], kde=True, bins=50)
plt.title('Lead Time Distribution')
plt.xlabel('Lead Time (days)')
plt.show()

# -------------------------------------------------------
# Interpretation:
# The distribution of lead_time is highly right-skewed.
# Most bookings occur within the first 100 days before check-in,
# with a long tail extending beyond 400 days.
#
# Despite this skew, logistic regression does not require
# predictors to be normally distributed — so we can proceed
# with raw lead_time as a covariate.
#
# Optional: If needed later, we can apply a log-transform using
np.log1p().
# -------------------------------------------------------
```

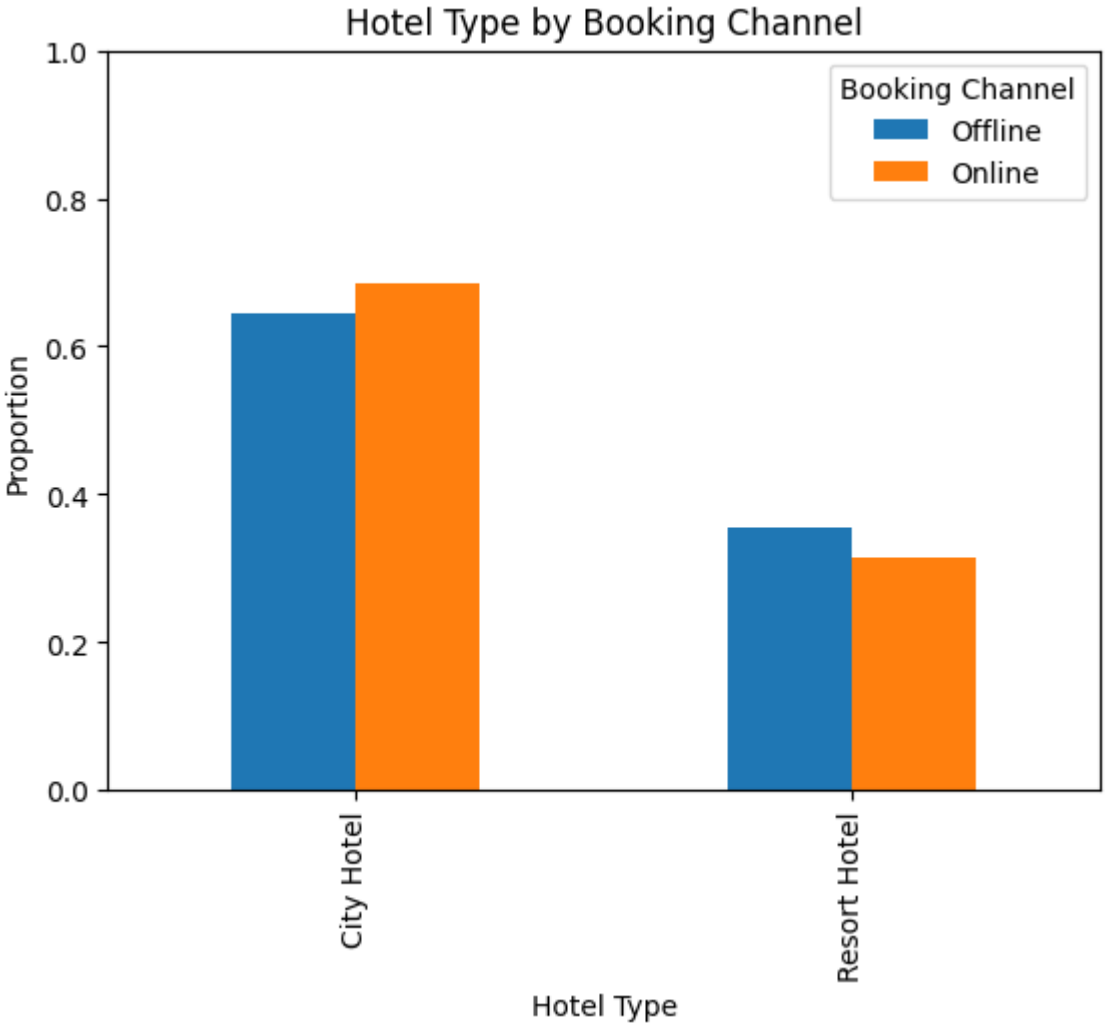

Lead Time Distribution by Booking Channel
Lead Time (days)
0
50
100
150
200
250
300
350
400
Offline
Online
Booking Channel


Lead Time Distribution
Count
0
5000
10000
15000
20000
25000
0
100
200
300
400
500
600
700
Lead Time (days)

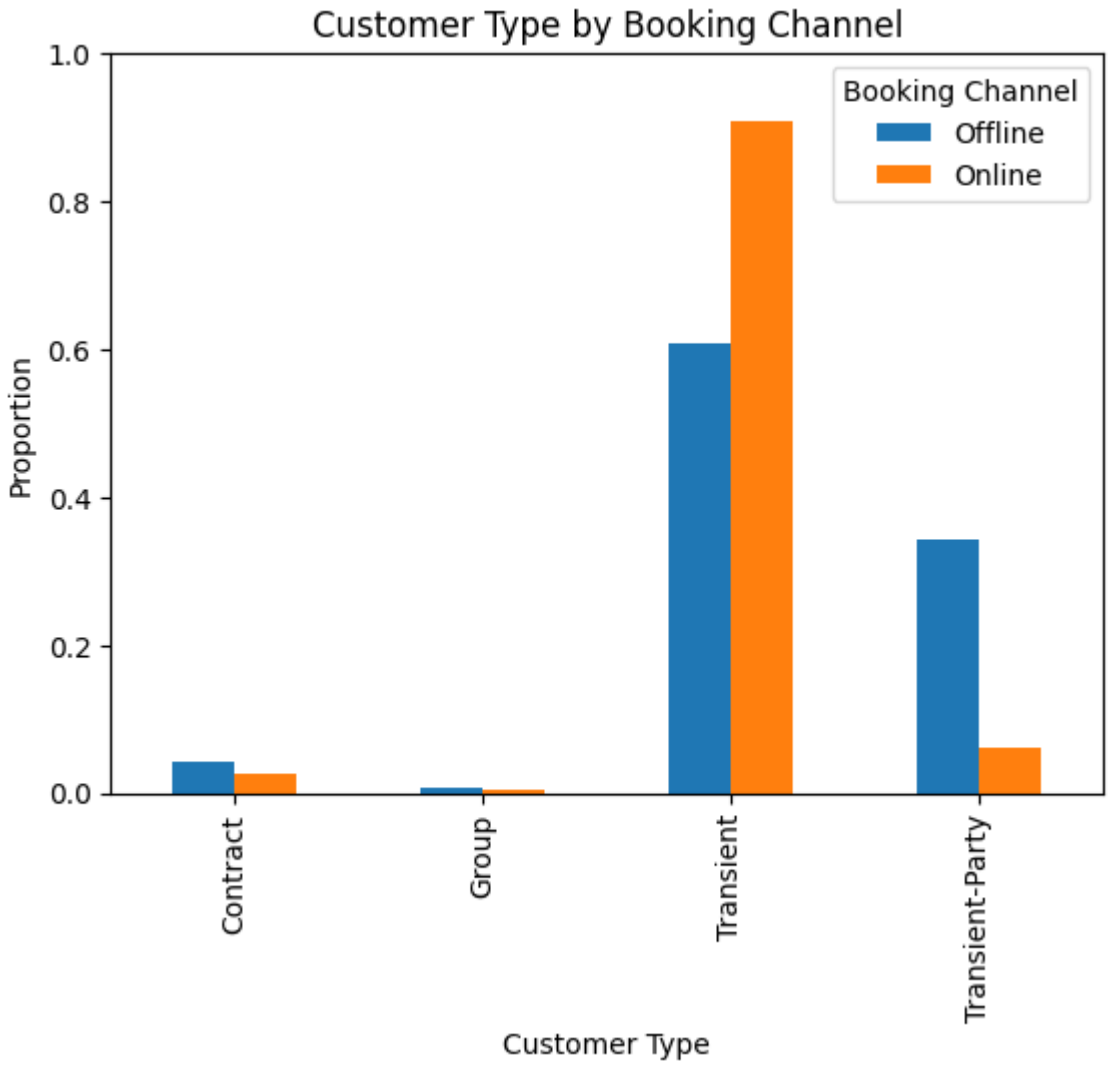
In [ ]:
```
# ============================================================
# Step 5B: Covariate Balance — Deposit Type (Side-by-Side)
# ============================================================

# Cross-tab: proportion of deposit types by booking_channel
deposit_crosstab = pd.crosstab(df['deposit_type'], df['booking_channel
'], normalize='columns')

# Plot deposit type distribution by channel (side-by-side bars)
deposit_crosstab.plot(kind='bar')
plt.title('Deposit Type by Booking Channel')
plt.ylabel('Proportion')
plt.xlabel('Deposit Type')
plt.legend(title='Booking Channel')
plt.ylim(0, 1)
plt.show()

# -------------------------------------------------------
# Interpretation:
# 'Non Refund' deposits are far more common in Offline bookings (~23%
vs <0.1%).
# This is a strong confounder: customers who can't cancel won't
cancel,
# regardless of booking method. We'll control for this.
# -------------------------------------------------------
```



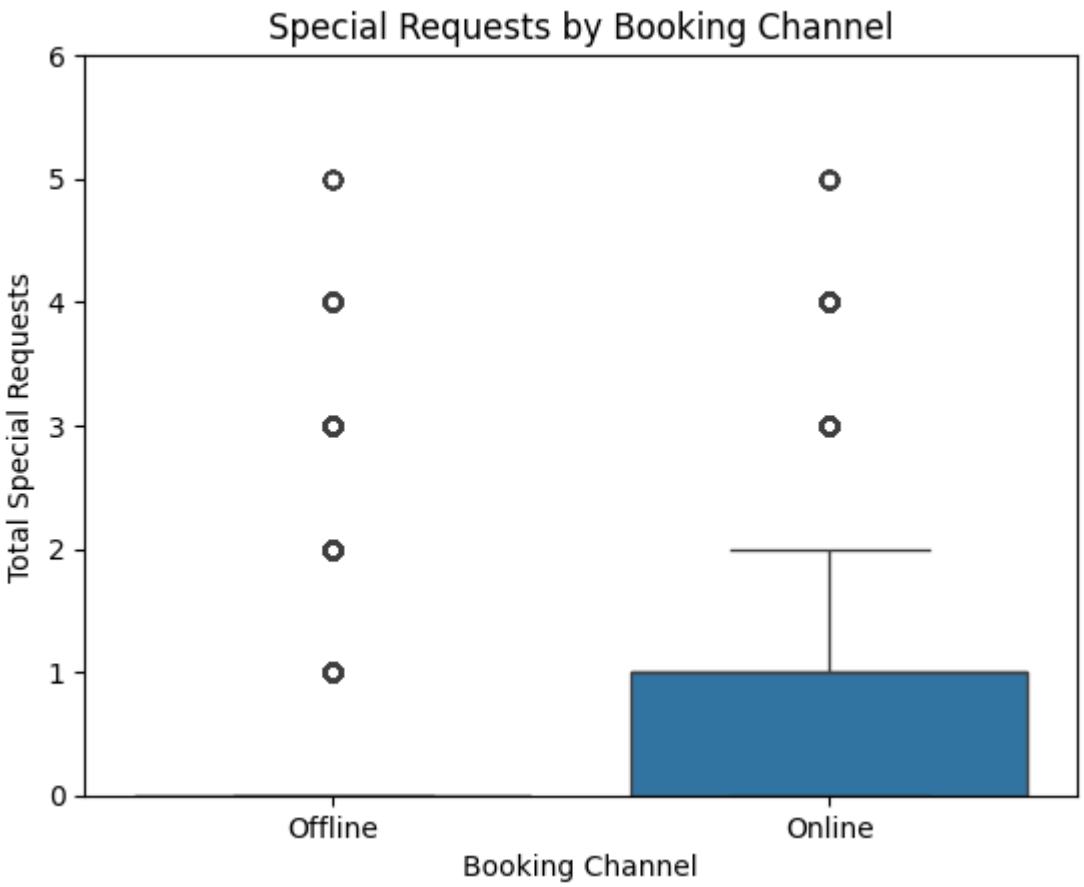
In [ ]:
```
# ============================================================
# Step 5C: Covariate Balance — Hotel Type (Side-by-Side)
# ============================================================

# Cross-tab: proportion of hotel types by booking_channel
hotel_crosstab = pd.crosstab(df['hotel'], df['booking_channel'], norm
alize='columns')

# Plot hotel type distribution by channel (side-by-side bars)
hotel_crosstab.plot(kind='bar')
plt.title('Hotel Type by Booking Channel')
plt.ylabel('Proportion')
plt.xlabel('Hotel Type')
plt.legend(title='Booking Channel')
plt.ylim(0, 1)
plt.show()

# -------------------------------------------------------
# Interpretation:
# City vs Resort hotel balance is similar across channels (~68% City
for Online vs ~64.5% Offline).
# This small imbalance may still be worth controlling for,
# in case hotel type moderates cancellation behavior.
# -------------------------------------------------------
```



In [ ]:
```
# ==============================================================
# Step 5D: Covariate Balance — Customer Type Distribution
# ==============================================================

# Compare distribution of customer_type across booking channels
customer_crosstab = pd.crosstab(df['customer_type'], df['booking_chan
nel'], normalize='columns')

# Side-by-side bar plot
customer_crosstab.plot(kind='bar')
plt.title('Customer Type by Booking Channel')
plt.ylabel('Proportion')
plt.xlabel('Customer Type')
plt.legend(title='Booking Channel')
plt.ylim(0, 1)
plt.show()

# -------------------------------------------------------
# This covariate reflects guest intent (e.g., Group vs Transient).
# If one booking channel has more 'Group' or 'Contract' bookings,
# that could affect cancellation behavior. We'll evaluate imbalance.
# -------------------------------------------------------

# -------------------------------------------------------
# Interpretation:
# Online bookings are overwhelmingly 'Transient' (~89%),
# while Offline bookings have a more diverse mix:
# ~61% Transient and ~35% Transient-Party.
# This imbalance could reflect different guest profiles
# (e.g., solo vs family/group travelers) and potentially
# impact cancellation behavior. We will control for it.
# -------------------------------------------------------
```


Customer Type by Booking Channel
Booking Channel
Offline
Online
Proportion
1.0
0.8
0.6
0.4
0.2
0.0
Contract
Group
Transient
Transient-Party
Customer Type

In [ ]:
```
# ==================================================================
# Step 5E: Covariate Balance — Total Special Requests Distribution
# ==================================================================

import seaborn as sns
import matplotlib.pyplot as plt

# Create boxplot of total special requests across booking channels
sns.boxplot(x='booking_channel', y='total_of_special_requests', data=d
f)
plt.title('Special Requests by Booking Channel')
plt.xlabel('Booking Channel')
plt.ylabel('Total Special Requests')
plt.ylim(0, df['total_of_special_requests'].max() + 1)
plt.show()

# -------------------------------------------------------
# Guests with more special requests are likely more committed.
# This behavioral covariate may reduce cancellation probability.
# We'll assess if it differs by booking channel and control for it.
# -------------------------------------------------------

# -------------------------------------------------------
# Interpretation:
# Online bookings show slightly higher total special requests
# than Offline bookings (median = 1 vs 0).
# Special requests may reflect customer engagement or commitment,
# and are expected to reduce cancellation probability.
# We will include this as a numeric covariate in our model.
# -------------------------------------------------------
```



In [ ]:
```python
# ============================================================
# Step 6: Logistic Regression with Covariate Adjustment
# ============================================================

import statsmodels.formula.api as smf
import numpy as np
import pandas as pd

# -------------------------------------------------------
# Convert relevant variables to categorical for modeling
# -------------------------------------------------------
df['booking_channel'] = df['booking_channel'].astype('category')
 # Treatment
df['deposit_type'] = df['deposit_type'].astype('category')
 # Refund policy
df['hotel'] = df['hotel'].astype('category')
 # City vs Resort
df['customer_type'] = df['customer_type'].astype('category')
 # Guest intent

# -------------------------------------------------------
# Fit logistic regression model
# Outcome: is_canceled (binary)
# Predictors: booking_channel (treatment),
# plus covariates: lead_time, deposit_type, hotel,
# customer_type, total_of_special_requests
# -------------------------------------------------------

model = smf.logit(
    "is_canceled ~ booking_channel + lead_time + deposit_type + hotel
+ customer_type + total_of_special_requests",
    data=df
).fit()

# View model summary
model.summary()

# -------------------------------------------------------
# Interpretation:
# Coefficients are in log-odds scale.
# We'll convert them to odds ratios next.
# -------------------------------------------------------
```

```
Optimization terminated successfully.
         Current function value: 0.473881
         Iterations 9
```

Out[ ]: Logit Regression Results

Dep. Variable:	is_canceled	**No. Observations:**	119386
Model:	Logit	**Df Residuals:**	119376
Method:	MLE	**Df Model:**	9
Date:	Fri, 09 May 2025	**Pseudo R-squ.:**	0.2811
Time:	05:05:05	**Log-Likelihood:**	-56575.
converged:	True	**LL-Null:**	-78695.
Covariance Type:	nonrobust	**LLR p-value:**	0.000

	coef	std err	z	P>|z|	[0.025	0.975]
Intercept	-1.8864	0.046	-41.041	0.000	-1.977	-1.796
booking_channel[T.Online]	1.2666	0.018	68.761	0.000	1.231	1.303
deposit_type[T.Non Refund]	5.7886	0.105	54.958	0.000	5.582	5.995
deposit_type[T.Refundable]	-0.0846	0.198	-0.427	0.669	-0.473	0.304
hotel[T.Resort Hotel]	-0.1878	0.016	-12.017	0.000	-0.218	-0.157
customer_type[T.Group]	-0.6456	0.150	-4.318	0.000	-0.939	-0.353
customer_type[T.Transient]	0.3464	0.045	7.771	0.000	0.259	0.434
customer_type[T.Transient-Party]	-0.0320	0.047	-0.685	0.493	-0.124	0.060
lead_time	0.0049	7.97e-05	61.436	0.000	0.005	0.005
total_of_special_requests	-0.7295	0.011	-66.340	0.000	-0.751	-0.708

In [ ]:
```
# ============================================================
# Convert coefficients to odds ratios for interpretability
# ============================================================

# Odds ratios from model coefficients
odds_ratios = np.exp(model.params)

# Confidence intervals for odds ratios
conf_int = model.conf_int()
conf_int_exp = np.exp(conf_int)
conf_int_exp.columns = ['CI Lower (2.5%)', 'CI Upper (97.5%)']

# Combine results into a summary table
odds_table = pd.concat([odds_ratios, conf_int_exp], axis=1)
odds_table.columns = ['Odds Ratio', 'CI Lower (2.5%)', 'CI Upper
(97.5%)']

# Display the table
display(odds_table)

# ------------------------------------------------------------
# Interpretation notes:
# Odds Ratio > 1 → Increases cancellation odds
# Odds Ratio < 1 → Decreases cancellation odds
# If CI crosses 1 → Not statistically significant
# ------------------------------------------------------------
```

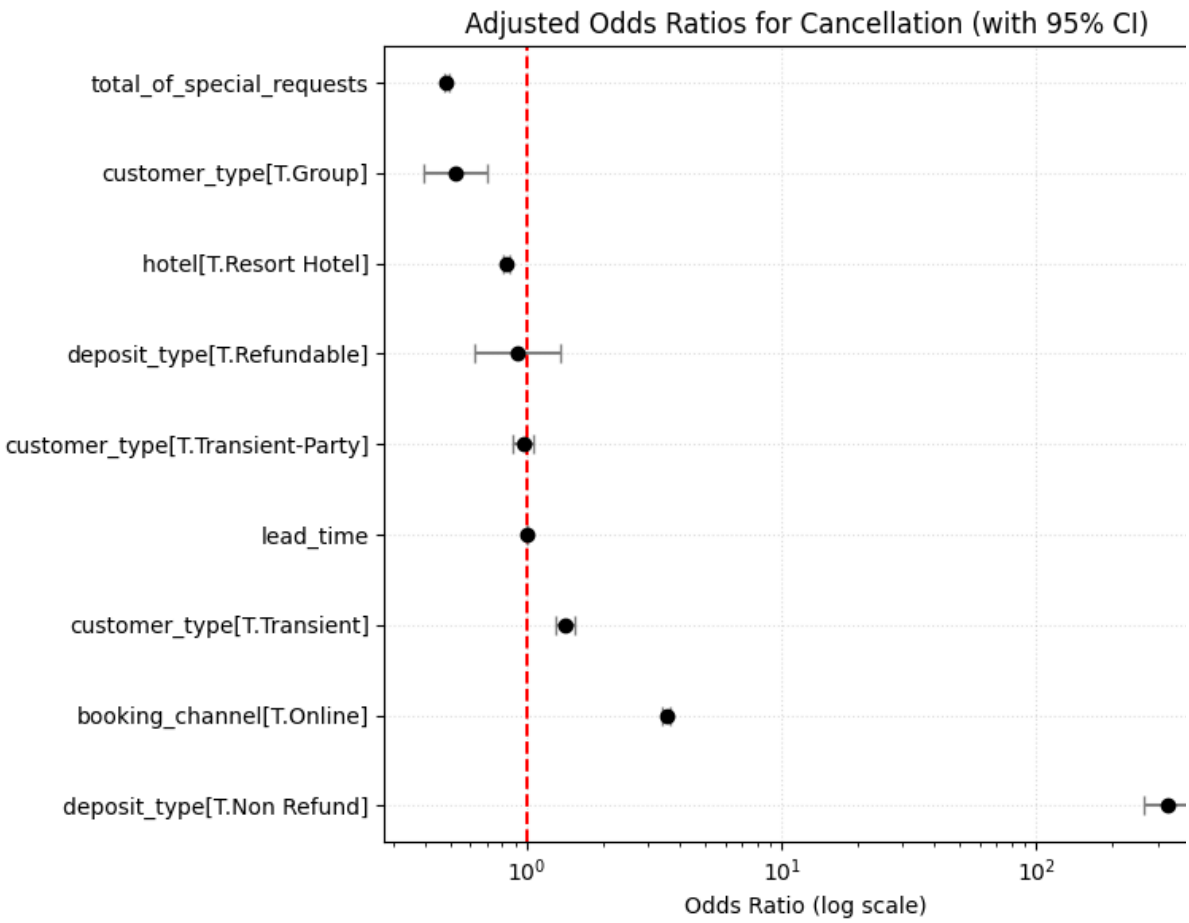
	Odds Ratio	CI Lower (2.5%)	CI Upper (97.5%)
Intercept	0.151609	0.138548	0.165902
booking_channel[T.Online]	3.548922	3.423076	3.679395
deposit_type[T.Non Refund]	326.561654	265.649679	401.440402
deposit_type[T.Refundable]	0.918876	0.623158	1.354926
hotel[T.Resort Hotel]	0.828792	0.803793	0.854569
customer_type[T.Group]	0.524342	0.391151	0.702885
customer_type[T.Transient]	1.413958	1.295664	1.543053
customer_type[T.Transient-Party]	0.968508	0.883810	1.061323
lead_time	1.004910	1.004753	1.005067
total_of_special_requests	0.482156	0.471876	0.492660

In [ ]:
```
# ================================================================
# Step 6B: Progressive Models to Compare Booking Channel Effect
# ================================================================

import statsmodels.formula.api as smf
import numpy as np
import pandas as pd

# ----------------------------------------------
# Fit Model 1: Booking Channel only (unadjusted)
# ----------------------------------------------
model1 = smf.logit("is_canceled ~ booking_channel",
data=df).fit(disp=0)

# ----------------------------------------------
# Fit Model 2: + lead_time
# ----------------------------------------------
model2 = smf.logit("is_canceled ~ booking_channel + lead_time", data=d
f).fit(disp=0)

# ----------------------------------------------
# Fit Model 3: + deposit_type
# ----------------------------------------------
model3 = smf.logit("is_canceled ~ booking_channel + lead_time +
deposit_type", data=df).fit(disp=0)

# ----------------------------------------------------
# Extract odds ratio and CI for booking_channel[T.Online]
# ----------------------------------------------------
def extract_booking_channel_effect(model):
    or_val = np.exp(model.params['booking_channel[T.Online]'])
    ci_low, ci_high = np.exp(model.conf_int().loc['booking_channel[T.
Online]'])
    return pd.Series({'Odds Ratio': or_val, 'CI Lower': ci_low, 'CI
Upper': ci_high})

results_table = pd.DataFrame({
    'Model 1\n(Booking Channel Only)':
extract_booking_channel_effect(model1),
    'Model 2\n+ lead_time': extract_booking_channel_effect(model2),
    'Model 3\n+ deposit_type': extract_booking_channel_effect(model3),
}).T

# Display comparison
display(results_table)

# ----------------------------------------------------
# Interpretation:
# Track how the odds ratio for booking_channel shifts
# across models. If it remains stable, the effect is robust.
# ----------------------------------------------------
```

	Odds Ratio	CI Lower	CI Upper
Model 1\n(Booking Channel Only)	0.974330	0.951670	0.997530
Model 2\n+ lead_time	1.260280	1.228991	1.292366
Model 3\n+ deposit_type	2.781078	2.700174	2.864405

In [ ]:

```
# ==============================================================
# Step 7A: Visual Summary — Plot Odds Ratios from Final Model
# ==============================================================

import matplotlib.pyplot as plt

# Reset index for clean plotting
plot_data = odds_table.reset_index()
plot_data = plot_data[plot_data['index'] != 'Intercept']  # Skip
intercept

# Sort descending by Odds Ratio
plot_data.sort_values(by='Odds Ratio', ascending=False, inplace=True)

# Set figure size
plt.figure(figsize=(8, 6))

# Plot odds ratios with CI error bars
plt.errorbar(
    x=plot_data['Odds Ratio'],
    y=plot_data['index'],
    xerr=[
        plot_data['Odds Ratio'] - plot_data['CI Lower (2.5%)'],
        plot_data['CI Upper (97.5%)'] - plot_data['Odds Ratio']
    ],
    fmt='o',
    color='black',
    ecolor='gray',
    capsize=4
)

# Add vertical line at OR = 1
plt.axvline(1, color='red', linestyle='--')

# Customize labels
plt.xlabel('Odds Ratio (log scale)')
plt.title('Adjusted Odds Ratios for Cancellation (with 95% CI)')
plt.xscale('log')
plt.tight_layout()
plt.grid(True, linestyle=':', alpha=0.4)
plt.show()

# -------------------------------------------------------------
# Interpretation:
# - Points right of 1 increase cancellation odds
# - Points left of 1 reduce them
# - booking_channel[T.Online], deposit_type[T.Non Refund],
#   and special_requests have the strongest effects.
# -------------------------------------------------------------




#Key Observations:
#deposit_type[T.Non Refund] has the largest effect by far (OR ≈ 326):
# Cancelling is extremely unlikely under non-refundable terms.

#booking_channel[T.Online]: OR ≈ 3.55
# After adjustment, online bookings are over 3.5x more likely to be
canceled than offline bookings.
```


Adjusted Odds Ratios for Cancellation (with 95% CI)
total_of_special_requests
customer_type[T.Group]
hotel[T.Resort Hotel]
deposit_type[T.Refundable]
customer_type[T.Transient-Party]
lead_time
customer_type[T.Transient]
booking_channel[T.Online]
deposit_type[T.Non Refund]
10^0
10^1
10^2
Odds Ratio (log scale)

In [ ]:
```
# ============================================================
# Step 8A: Power Analysis for Two-Proportion Z-Test
# ============================================================

from statsmodels.stats.power import zt_ind_solve_power

# Observed proportions
p1 = 0.3672  # Online
p2 = 0.3733  # Offline

# Pooled standard deviation estimate
effect_size = abs(p1 - p2) / np.sqrt((p1 * (1 - p1) + p2 * (1 - p2)) /
2)

# Sample sizes
n1 = 56476
n2 = 62910
n_total = 1 / (1/n1 + 1/n2)  # Harmonic mean for unequal sizes

# Power calculation (two-sided test, alpha=0.05)
power = zt_ind_solve_power(effect_size=effect_size, nobs1=n_total, al
pha=0.05, alternative='two-sided')

print(f"Post Hoc Power: {power:.4f}")

#“Given the observed difference in cancellation rates between Online
and Offline groups, do we have enough power to detect that
difference?”

#With over 119,000 observations, the power to detect such a small
difference (~0.6%) is only 33.8%, which is well below the conventional
80% threshold.
#This means  study is underpowered to reliably detect such a small
difference using a two-proportion test alone.
```

```
Post Hoc Power: 0.3379
```

In [ ]:
```python
# ============================================================
# Step 8B: What minimum effect size can we detect?
# ============================================================

# Solve for minimum detectable effect size given 80% power
min_effect = zt_ind_solve_power(
    effect_size=None,
    nobs1=n_total,
    alpha=0.05,
    power=0.80,
    alternative='two-sided'
)

# Convert back to raw difference (approximate)
avg_p = (p1 + p2) / 2
min_diff = min_effect * np.sqrt(2 * avg_p * (1 - avg_p))

print(f"Minimum detectable difference at 80% power: {min_diff:.4f}")

#“What is the minimum difference in cancellation rates I can detect at
80% power?”

# Current sample size is sufficient to detect differences in
cancellation rates of 1.57% or larger.
#Since  actual observed difference is only 0.61%, it falls below the
detection threshold — explaining why the two-proportion Z-test was not
statistically significant, even with a large sample.

#However,  logistic regression model (adjusted for confounders)
revealed a much stronger treatment effect — validating your
experimental insight.
```

```
Minimum detectable difference at 80% power: 0.0157
```